A Stochastic-Optimal Control Design for Hybrid Switched CSTR Systems

***Abstract*— Recent advances have focused heavily on developing optimal control strategies for switched Continuous Stirred Tank Reactor (CSTR) systems. While these approaches often perform well under ideal conditions, their reliability tends to falter in real-world applications where disturbances, process and measurement noises are inevitable. This paper proposes an observer design by combining the Extended Kalman Filter (EKF) with an Extended State Observer (ESO), that when combined with switching and control policies derived from the Hamilton-Jacobi-Bellman (HJB) equation, significantly improves system stability and performance under non-ideal conditions. Simulation results illustrate the effectiveness of the proposed method and its potential for real-world deployment in chemical process control.**

Keywords—Optimal control, CSTR, Optimal Estimation, Monte Carlo, Extended State Observer.

I. INTRODUCTION

The switched system is a holistic system consisting of multiple sub systems with diverse dynamics, where one and only one subsystem is active simultaneously. The hybrid control of the switched system is to design a switching law to activate subsystems intermittently in a time horizon and give a control input to regulate the system state. When given a cost function associated with the state and control, the problem escalates to the concept of optimal control. Compared with systems with single dynamics, the research on the switched system involves the selection of subsystems, the determination of switching times, the input design, etc., making the problem theoretically and computationally tricky [1].

In chemical engineering, the Continuous Stirred Tank Reactor (CSTR) is a fundamental flow reactor design. Reactants continuously enter the vessel, where vigorous stirring by an impeller ensures thorough mixing. CSTRs offer significant operational advantages, including simplicity, inherent stability, and the ability to sustain steady production. Designing and operating a CSTR effectively requires careful management of key parameters such as flow rate, temperature, and reactant concentration. Given the reactor's widespread use across the chemical industry, numerous strategies have been developed to optimize its performance and control its operation [2].

The work done in [2] formulate the optimal control problem for switched CSTR systems as a (HJB) equation over a family of affine nonlinear subsystems. Unlike earlier approaches that fix the number or order of switches in advance, their formulation allows both the sequence of subsystem activations and the timing of switches to emerge directly from the minimization of the Hamiltonian at each instant. By proving that the Hamiltonian is concave with respect to the switching signal, they show that, at every time step, the optimal active subsystem is the one whose local Hamiltonian contribution is minimal. To overcome the fundamental difficulty that the HJB equation generally does not admit closed-form solutions, the authors employ Galerkin's spectral method [3]. This approach approximates the unknown value function using a truncated polynomial basis and projects the HJB residual onto the same basis, yielding a set of algebraic equations for the basis coefficients. Compared with neural-network-based actor–critic schemes, Galerkin's method offers a more transparent convergence analysis, lighter computational load, and reduced risk of entrapment in local optima.

Although the method proposed in [2] demonstrates desirable performance under ideal conditions, it encounters challenges when confronted with factors inevitable in real-world applications. Fundamentally, the sensitivity of the switching mechanism proposed in [2] to measurement noise, process noise, or disturbances can degrade response quality and, in some cases, may even lead to system instability.

The Kalman Filter (KF) provides an optimal recursive method for state estimation in dynamical systems by explicitly modeling both process and measurement noise. Its process begins with a prediction phase, where the system's next state and uncertainty are projected using the process model and noise covariance. Upon receiving new measurements, the update phase calculates a Kalman Gain that optimally weights the prediction against incoming sensor data based on their relative uncertainties. This mechanism automatically adapts to noise levels: high process noise increases reliance on measurements, while high measurement noise prioritizes model predictions. By continuously fusing forecasts and observations in a minimum-variance framework, the KF effectively minimizes the impact of process unpredictability and sensor inaccuracies, yielding robust real-time state estimates [4].

Extended State Observers (ESOs) offer a powerful and unified strategy for handling unknown disturbances in dynamic systems. Their key innovation lies in augmenting the system's state vector to include a new component as an extended state, that represents the total disturbance acting on the system. This includes not only external forces or noise, but also unmodeled dynamics and parameter variations. Unlike conventional observers, which aim to reconstruct system states based purely on known inputs and models, ESOs treat these unknowns as part of the state and estimate them in real time [5].

This paper proposes a novel optimal estimator that synthesizes Kalman Filter (KF) principles to mitigate measurement and process noise and an Extended State Observer (ESO) to estimate disturbances. When integrated with the optimal control law and switching strategy from [2], this approach significantly attenuates adverse factors in practical implementations.

II. Problem Formulation

We consider a constant-volume (CSTR) with multiple inlet streams, each constituting a distinct subsystem characterized by unique reactant parameters. A switching valve selects a single active inlet stream at any time, defining the operative subsystem. Within the reactor, an exothermic irreversible reaction $A \rightarrow B$ occurs. The system is cooled by a coolant stream with constant flow rate but adjustable temperature [2]. Consider the system model given in [6]:

$$\dot{C}_A = \frac{q_i}{V}\left(C_{Afi} - C_A\right) - a_0 \exp\left(-\frac{E}{RT}\right) C_A \tag{1a}$$

$$\dot{T} = \frac{q_i}{V}\left(T_{fi} - T\right) - a_1 \exp\left(-\frac{E}{RT}\right) C_A + a_2(T_C - T) \tag{1b}$$

Where a_0, a_1 and a_2 are constants, C_A is concentration of reactant A, C_{Af} is feed concentration of reactant A, E is activation energy, q is feed flow rate, R is gas constant, T is reactor temperature, T_C is coolant temperature, T_f is feed temperature, V is volume of the reactor and i represents the subsystems. By considering $T_C{}^*$, $C_A{}^*$ and T^* as nominal operating condition, the systems states and control input can be considered as $x_1 = C_A - C_A{}^*$, $x_2 = T - T^*$ and $u = T_C - T_C{}^*$. Now the chemical process can be described by:

$$\dot{x}_1 = \frac{q_i}{V}\left(C_{Afi} - C_A{}^* - x_1\right) - a_0 \exp\left(-\frac{E/R}{x_2 + T^*}\right)(x_1 + C_A{}^*) \tag{2a}$$

$$\dot{x}_2 = \frac{q_i}{V}\left(T_{fi} - T^* - x_2\right) - a_1 \exp\left(-\frac{E/R}{x_2 + T^*}\right)(x_1 + C_A{}^*) + a_2(T_C{}^* - T^* - x_2) + a_2 u \tag{2b}$$

The control objectives are to Dynamically select the optimal inlet stream (subsystem activation) and regulate the systems states while minimizing the cost function:

$$J(x_0, u) = \int_0^\infty (x^T Q x + r u^2) dt \tag{3}$$

Where $x = [x_1 \quad x_2]^T$ is state vector, Q is a positive definite matrix and r is a positive scalar.

Each subsystem can be modeled as a time-invariant affine nonlinear system, as defined below:

$$\dot{x}(t) = f_i\big(x(t)\big) + g_i\big(x(t)\big) u\big(x(t)\big) \tag{4}$$

The system is formulated by incorporating the parameter vector v as a switching signal as follows:

$$\dot{x}(t) = \sum_{i=1}^{2} v_i \left[f_i\big(x(t)\big) + g_i\big(x(t)\big) u\big(x(t)\big)\right] \tag{5}$$

Where $v_i \in \{0,1\}$ and the vector $v = [v_1 \quad v_2]^{\mathrm{T}}$ is defined accordingly. By considering switched nonlinear model of (5) and the cost function described by (3), According to the principle of optimality, this leads to a unique optimal value function $V^*(x)$, which satisfies the HJB equation tailored for switched systems, expressed as follows:

$$\frac{\partial V^*}{\partial t} + \min_u \{H\} = 0 \tag{6}$$

Where H denotes the Hamiltonian, defined as:

$$H = x^T Q x + r u^2 + \frac{\partial V^*}{\partial x} \sum_{i=1}^{2} v_i \left[f_i\big(x(t)\big) + g_i\big(x(t)\big) u\big(x(t)\big)\right] \tag{7}$$

Based on the assumptions and proofs provided in [2], the optimal control law and switching law can be derived as follows:

$$u^* = -\frac{1}{2} r^{-1} g_i^T(x) \frac{\partial^T V_i{}^*}{\partial x} \tag{8}$$

$$i(t) = \operatorname*{argmin}_{i=1,2} \left\{\frac{\partial V_i{}^*}{\partial x} f_i(x) + \frac{1}{2}\frac{\partial V_i{}^*}{\partial x} g_i(x) u^*\right\} \tag{9}$$

Where V_i^* represents the optimal value function of the ith subsystem.

While the switching laws derived in the previous section facilitate optimal system behavior, a key challenge arises due to the inaccessibility of the term $\frac{\partial V^*}{\partial x}$. For general nonlinear optimal control problems, obtaining a closed-form analytical expression for the optimal value function is often intractable or outright impossible. This limitation renders discriminant (8) and (9) unusable. Consequently, a widely adopted approach is to design algorithms that approximate the value function. In this paper we apply Galerkin's spectral method for approximating the optimal value function and control law [3].

To this end, we assume that the function $V_i(x)$ can be represented as an infinite series expansion over a set of known, continuous, and linearly independent basis functions $\varphi_k(x)$, as follows:

$$V_i(x) = \sum_{k=1}^{\infty} c_k^i \varphi_k(x) \tag{10}$$

Where c_k^i are coefficients to be determined. It is further assumed that V_i can be approximated to arbitrary accuracy by taking a finite truncation of the series. The truncation representation of V_i using the first M basis functions is given by:

$$V_{i,M}(x) = \sum_{k=1}^{M} c_k^i \varphi_k(x) = \Phi_M^T C_M^i \tag{11}$$

where $\Phi_M = [\varphi_1 \quad \cdots \quad \varphi_M]^T$ is the vector of basis functions, and $C_M^i = [c_1^i \quad \cdots \quad c_M^i]^T$ is the corresponding coefficient vector. By defining the inner product as follows:

$$\langle \mu, \rho \rangle_\Omega = \int_\Omega \mu(x)\, \rho(x)\, dx \tag{12}$$

Where μ and ρ are scalar functions defined on the domain of integration Ω. The notation $\langle \Phi_M, \Phi_M^T \rangle$ denotes an $M \times M$ matrix whose ijth entry corresponds to the inner product $\langle \varphi_i, \varphi_j \rangle$. Similarly, $\langle \Phi_M, \rho \rangle$ represents an $M \times 1$ vector in which the ith entry is given by $\langle \varphi_i, \rho \rangle$. Utilizing the aforementioned operators, Galerkin's spectral method formulation as detailed in algorithm 1 can be derived accordingly [3].

To mitigate the uncertainty caused by unscheduled switching, the system enforces a fixed time interval τ between potential switching events. This ensures that switching decisions are evaluated only at predetermined, uniformly spaced time instances, rather than continuously [2].

As previously stated, although the proposed method provides a closed-form control law, the switching rule remains highly sensitive to undesired noises and disturbances. The following section addresses this performance limitation by introducing an optimal estimator.

III. Optimal Estimator Design

As it's clear for implementing optimal control law (8) and selecting the optimal subsystem based on (9) knowing states values is vital. So, we developed EKF for estimating states for hybrid system in presence of input disturbance, process and measurement noises. For completeness we express the algorithm as it has been introduced in [6].

Algorithm 1 Galerkin's Spectral Method.

Input: $\boldsymbol{f, g, Q, r, u_0, \Phi_M, \varepsilon, \Omega}$

Calculate:

$\langle \boldsymbol{\Phi_M}, \boldsymbol{x^T Q x} \rangle$, $\langle \boldsymbol{\Phi_M}, (\boldsymbol{\nabla \Phi_M f})^{\boldsymbol{T}} \rangle$

$\langle \boldsymbol{\Phi_M}, \boldsymbol{r u_0}^{\boldsymbol{2}} \rangle$, $\langle \boldsymbol{\Phi_M}, (\boldsymbol{\nabla \Phi_M g u_0})^{\boldsymbol{T}} \rangle$

$\boldsymbol{A} = \langle \boldsymbol{\Phi_M}, (\boldsymbol{\nabla \Phi_M f})^{\boldsymbol{T}} \rangle + \langle \boldsymbol{\Phi_M}, (\boldsymbol{\nabla \Phi_M g u_0})^{\boldsymbol{T}} \rangle$

$\boldsymbol{b} = -\langle \boldsymbol{\Phi_M}, \boldsymbol{x^T Q x} \rangle - \langle \boldsymbol{\Phi_M}, \boldsymbol{r u_0}^{\boldsymbol{2}} \rangle$

Set: $\boldsymbol{n = 0}$

Iterate:

$\boldsymbol{C_M^n = A^{-1} b}$, $\frac{\boldsymbol{\partial V_n}}{\boldsymbol{\partial x}} = (\boldsymbol{C_M^n})^{\boldsymbol{T}} \boldsymbol{\nabla \Phi_M}$

$$\boldsymbol{u_{n+1}} = -\frac{\boldsymbol{1}}{\boldsymbol{2}} \boldsymbol{r^{-1} g^T} \frac{\boldsymbol{\partial^T V_n}}{\boldsymbol{\partial x}}$$

$$\boldsymbol{L_n} = \sum_{\boldsymbol{k=1}}^{\boldsymbol{M}} \boldsymbol{c_k^n} \langle \boldsymbol{\Phi_M}, \boldsymbol{\nabla \varphi_k g r^{-1} g^T \nabla \Phi_M^T} \rangle$$

$$\boldsymbol{A} = \langle \boldsymbol{\Phi_M}, (\boldsymbol{\nabla \Phi_M f})^{\boldsymbol{T}} \rangle - \frac{\boldsymbol{1}}{\boldsymbol{2}} \boldsymbol{L_n}$$

$$\boldsymbol{b} = -\langle \boldsymbol{\Phi_M}, \boldsymbol{x^T Q x} \rangle - \frac{\boldsymbol{1}}{\boldsymbol{4}} \boldsymbol{L_n C_M^n}$$

$\boldsymbol{n = n + 1}$

End if: $\left\| \boldsymbol{C_M^{n+1} - C_M^n} \right\| \leq \boldsymbol{\varepsilon}$

Output: $\frac{\boldsymbol{\partial V^*}}{\boldsymbol{\partial x}} = \left(\boldsymbol{C_M^{n+1}}\right)^{\boldsymbol{T}} \boldsymbol{\nabla \Phi_M}, \boldsymbol{u^*} = -\frac{\boldsymbol{1}}{\boldsymbol{2}} \boldsymbol{r^{-1} g^T} \frac{\boldsymbol{\partial V^*}}{\boldsymbol{\partial x}}$

Consider the following nonlinear system:

$$\begin{cases} \dot{x}(t) = f(x, u, \omega, t) \\ y(t) = h(x, v, t) \end{cases} \tag{13}$$

Where $\omega(t) \sim (0, Q)$ and $v(t) \sim (0, R)$ are process and measurement noises.

Then we introduce following matrixes for implementing EKF as:

$$\begin{cases} A_i(t) = \left.\frac{\partial f_i}{\partial x}\right|_{\hat{x}} \\ L_i(t) = \left.\frac{\partial f_i}{\partial \omega}\right|_{\hat{x}} \\ C_i(t) = \left.\frac{\partial f_i}{\partial x}\right|_{\hat{x}} \\ M_i(t) = \left.\frac{\partial f_i}{\partial v}\right|_{\hat{x}} \\ \widetilde{R}_i = M_i R M_i^T \\ \widetilde{Q}_i = L_i Q L_i^T \end{cases} \tag{14}$$

Then we can use the following formulas to estimate the states of the system.

$$\dot{\hat{x}}(t) = f_i(\hat{x}, u, \omega_0, t) + K(y(t) - h_i(\hat{x}, v_0, t)) \tag{15}$$

$$K(t) = P(t) C_i^T \tilde{R}_i^{-1}(t) \tag{16}$$

$$\dot{P}(t) = A_i(t)P(t) + P(t)A_i^T(t) + \widetilde{Q}_i(t) - P(t)C_i^T(t)\tilde{R}_i^{-1}(t)C_i(t)P(t) \tag{17}$$

Where $\omega_0 = 0, v_0 = 0$ and we use a guess for initial condition for $\hat{x}(0)$ and $\hat{P}(0)$ in order to drive (15) to (17).

Remark 1: the index i in (14) to (17) indicates which subsystem have to be used in specific time in estimation.

IV. Stochastic Optimal Controller Design

For hybrid system (4) the optimal control is obtained by using optimal law for each subsystem with optimal switching law and this subject has been proved in [2] as mentioned before.

The proposed method in [2] was with no disturbance and it was pure deterministic. Here we used EKF in order to meet this shortcoming.

Considering the following nonlinear system:

$$\dot{x}(t) = f_i\big(x(t)\big) + g_i\big(x(t)\big)u\big(x(t)\big) + d(t) + \omega(t) \tag{18}$$

With the following measurement:

$$y(t) = I_{2\times 2}x(t) + v(t) \tag{19}$$

Where $d(t) = [0, d]^T, \omega(t) = [\omega_1(t), \omega_2(t)]^T, v(t) = [v_1(t), v_2(t)]^T, f_i\big(x(t)\big) = [f_{1i}(x), f_{2i}(x)]^T, g_i\big(x(t)\big) = [g_{1i}(x), g_{2i}(x)]^T$ and $x(t) = [x_1(t), x_2(t)]^T$.

For rejecting the stated disturbance we use ESO to estimate and compensate it. By putting $x_3(t) = d$ and then we have $\dot{x}_3(t) = 0$ we reach to the following extended state space:

$$\dot{x}_1(t) = f_{1i}\big(x(t)\big) + g_{1i}\big(x(t)\big)u\big(x(t)\big) + \omega_1(t) \tag{20a}$$

$$\dot{x}_2(t) = f_{2i}\big(x(t)\big) + g_{2i}\big(x(t)\big)u\big(x(t)\big) + x_3(t) + \omega_2(t) \tag{20b}$$

$$\dot{x}_3(t) = 0 \tag{20c}$$

Here we use EKF for estimating system (20) states and use the optimal control strategy as derived in [2] but instead of using the states we use the estimated states $\hat{x}(t)$.

We define control law as $u(t) = u^*(t) - \frac{1}{g_{2i}(x(t))}\hat{x}\_3(t)$ and $u^*(t)$ is defined in (8). And switching law will be the same as (9) but instead of using measurement we use the estimated system's states to mitigate the noises.

Remark 2: Here the control law that presented before will reject the step disturbance and the sufficient condition is that $g_{2i}\big(x(t)\big) \neq 0$ and according to the CSTR in (2) this condition is satisfied.

References

[1] F. Zhu and P. J. Antsaklis, "Optimal control of hybrid switched systems: A brief survey," *Discrete Event Dynamic Systems*, vol. 25, no. 3, pp. 345-364, 2015.